Supplementary Analysis: Cross-Model Explainability

ULB Credit Card Fraud Dataset

Arjun Sekhar

2026-10-04

1. Introduction

1.1. Objective

This notebook compares how LR, Cox, RF and RSF attribute predictive importance across the same ULB predictors. The Methodology Notebook (Section 9) explains RSF through permutation importance, SHAP and LIME, but provides no comparable view for the other three models. Here, all four are fitted on identical subsamples and their importance rankings are compared directly.

1.2. Motivation

Supervisor review (October 2026) asked how the RSF explanations compare with LR, RF and Cox. Since the four models produce different kinds of importance (standardised coefficients for LR and Cox, permutation importance for the forests), the comparison is made on rankings rather than raw magnitudes.

2. Libraries and Data

```
## Step 1: Load required packages (dplyr loaded last to avoid select() masking)
library(survival)
library(randomForestSRC)
library(glmnet)
library(caret)
library(ggplot2)
library(tidyr)
library(pander)
library(dplyr)
```

```
## Step 1: Set seed - same position as Methodology Notebook Section 3.1, so the
## downsampled working dataset below reproduces fraud_subset exactly
set.seed(44940076)

## Step 2: Load data
fraud_data <- read.csv("creditcard.csv", header = TRUE)
fraud_data <- na.omit(fraud_data)

## Step 3: Survival variables (Methodology Notebook Section 5.1)
fraud_data$time   <- 1:nrow(fraud_data)   # proxy time index
fraud_data$status <- fraud_data$Class      # event indicator

## Step 4: Downsampled working dataset (Methodology Notebook Section 5.2)
fraud_cases     <- subset(fraud_data, Class == 1)
nonfraud_cases  <- subset(fraud_data, Class == 0)
```

```r
nonfraud_sample <- nonfraud_cases[sample(nrow(nonfraud_cases), 50000), ]
fraud_subset    <- rbind(fraud_cases, nonfraud_sample)

dim(fraud_subset)
```

```
## [1] 50492    33
```

3. Design

Each of 30 repeated stratified 70-30 splits draws an importance subsample from the training partition, keeping all fraud cases and 10,000 non-fraud cases, as in the Methodology Notebook (Section 9.2). Predictors are standardised within each subsample so that LR and Cox coefficient magnitudes are comparable across predictors. All four models use the primary predictor set (V1-V28, Amount and raw `Time`):

- **LR:** absolute standardised coefficients
- **Cox (ridge):** absolute standardised coefficients at the cross-validated penalty
- **RF and RSF:** permutation importance (`importance = "random"`, 50 trees, as in Section 9.2)

```r
## Step 1: Predictor set and storage
pred_set  <- c(paste0("V", 1:28), "Amount", "Time")
n_splits  <- 30
imp_rows  <- vector("list", n_splits)
coef_rows <- vector("list", n_splits)

## Step 2: Set seed
set.seed(44940076)

## Step 3: Fit all four models on each split's importance subsample
for (i in 1:n_splits) {
  message("Explainability split ", i, " of ", n_splits, " - ", Sys.time())

  # 1. Stratified split, then importance subsample from the training partition
  idx     <- createDataPartition(fraud_subset$status, p = 0.7, list = FALSE)
  train_i <- fraud_subset[idx, ]
  fr      <- train_i[train_i$status == 1, ]
  nf      <- train_i[train_i$status == 0, ]
  sub     <- rbind(fr, nf[sample(nrow(nf), 10000), ])

  # 2. Standardise predictors within the subsample
  sub_s <- data.frame(scale(sub[, pred_set]),
                      status = sub$status,
                      time   = sub$time,
                      Class  = factor(sub$Class, levels = c(0, 1)))

  # 3. LR
  lr_fit  <- glm(reformulate(pred_set, response = "status"),
                 data = sub_s, family = binomial)
  lr_coef <- coef(lr_fit)[pred_set]

  # 4. Cox (ridge)
  cox_cv   <- cv.glmnet(as.matrix(sub_s[, pred_set]), Surv(sub_s$time, sub_s$status),
                        family = "cox", alpha = 0)
  cox_coef <- as.matrix(coef(cox_cv, s = "lambda.min"))[pred_set, 1]

  # 5. RF (classification), permutation importance
```

```
  rf_fit <- rfsrc(reformulate(pred_set, response = "Class"), data = sub_s,
                  ntree = 50, nodesize = 15, splitrule = "gini",
                  nthread = 4, fast = TRUE, importance = "random")
  rf_imp <- rf_fit$importance[pred_set, "all"]

  # 6. RSF, permutation importance
  rsf_fit <- rfsrc(as.formula(paste("Surv(time, status) ~", paste(pred_set, collapse = " + "))),
                   data = sub_s, ntree = 50, nodesize = 15, splitrule = "logrank",
                   nthread = 4, fast = TRUE, importance = "random")
  rsf_imp <- rsf_fit$importance[pred_set]

  # 7. Store
  imp_rows[[i]]  <- data.frame(split = i, feature = pred_set,
                               LR = abs(lr_coef), Cox = abs(cox_coef),
                               RF = rf_imp, RSF = rsf_imp)
  coef_rows[[i]] <- data.frame(split = i, feature = pred_set,
                               LR = lr_coef, Cox = cox_coef)
}

## Step 4: Save
saveRDS(list(imp  = do.call(rbind, imp_rows),
             coef = do.call(rbind, coef_rows)), "explain_ulb.rds")
```

4. Results

4.1. Importance Rankings by Model

```
## Step 1: Load results and average importance across splits
explain_ulb <- readRDS("explain_ulb.rds")
model_cols  <- c("LR", "Cox", "RF", "RSF")

imp_mean <- explain_ulb$imp %>%
  dplyr::group_by(feature) %>%
  dplyr::summarise(dplyr::across(dplyr::all_of(model_cols), mean), .groups = "drop")

## Step 2: Rank each predictor within each model (1 = most important)
rank_tab <- imp_mean %>%
  dplyr::mutate(dplyr::across(dplyr::all_of(model_cols), ~ rank(-.x), .names = "{.col}_rank")) %>%
  dplyr::arrange(RSF_rank)

pander(rank_tab %>% dplyr::select(feature, dplyr::ends_with("_rank")),
       caption = "Importance Rank by Model (1 = Most Important, 30 Splits)")
```

Table 1: Importance Rank by Model (1 = Most Important, 30 Splits)

feature	LR_rank	Cox_rank	RF_rank	RSF_rank
V14	2	3	1	1
V4	5	2	6	2
Time	20	1	23	3
V12	3	19	3	4
V17	1	26	4	5
V10	4	10	2	6

feature	LR_rank	Cox_rank	RF_rank	RSF_rank
V11	10	4	5	7
V19	23	20	19	8
Amount	16	11	16	9
V16	7	30	7	10
V7	6	12	9	11
V8	14	23	26	12
V20	22	25	24	13
V18	12	17	11	14
V3	8	22	8	15
V13	21	13	25	16
V21	19	29	13	17
V23	30	15	30	18
V15	24	14	28	19
V2	15	7	12	20
V28	27	18	18	21
V5	9	27	15	22
V9	13	21	10	23
V1	11	5	14	24
V25	29	8	22	25
V24	28	24	29	26
V26	26	6	27	27
V22	18	28	20	28
V27	25	9	17	29
V6	17	16	21	30

```
## Step 3: Top five predictors per model
top5 <- sapply(model_cols, function(m) imp_mean$feature[order(-imp_mean[[m]])][1:5])
pander(data.frame(Rank = 1:5, top5), caption = "Top Five Predictors by Model")
```

Table 2: Top Five Predictors by Model

Rank	LR	Cox	RF	RSF
1	V17	Time	V14	V14
2	V14	V4	V10	V4
3	V12	V14	V12	Time
4	V10	V11	V17	V12
5	V4	V1	V11	V17

```
## Step 4: Rank of raw Time in each model, for inline reporting
time_ranks <- rank_tab %>% dplyr::filter(feature == "Time") %>%
  dplyr::select(dplyr::ends_with("_rank"))
```

4.2. Agreement Between Models

```
## Step 1: Spearman rank correlation between models' importance profiles
rank_cor <- cor(imp_mean[, model_cols], method = "spearman")
pander(round(rank_cor, 3), caption = "Spearman Rank Agreement Between Models")
```

Table 3: Spearman Rank Agreement Between Models

	LR	Cox	RF	RSF
LR	1	-0.002	0.873	0.604
Cox	-0.002	1	0.083	0.187
RF	0.873	0.083	1	0.552
RSF	0.604	0.187	0.552	1

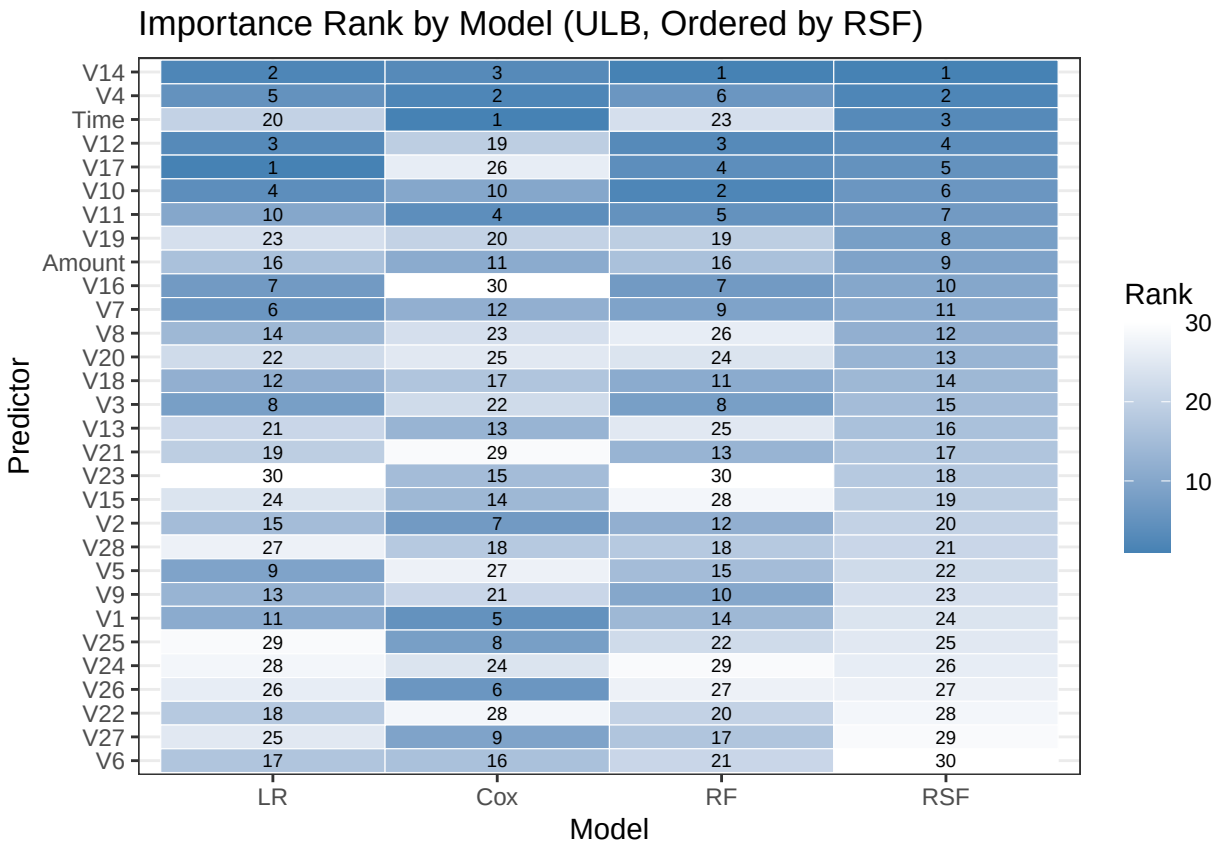
```
## Step 2: Top-five overlap between each pair of models
top5_overlap <- outer(model_cols, model_cols, Vectorize(function(a, b)
  length(intersect(top5[, a], top5[, b]))))
dimnames(top5_overlap) <- list(model_cols, model_cols)
pander(top5_overlap, caption = "Number of Shared Top-Five Predictors")
```

Table 4: Number of Shared Top-Five Predictors

	LR	Cox	RF	RSF
LR	5	2	4	4
Cox	2	5	2	3
RF	4	2	5	3
RSF	4	3	3	5

```
## Step 3: Rank heatmap
rank_long <- rank_tab %>%
  dplyr::select(feature, dplyr::ends_with("_rank")) %>%
  tidyr::pivot_longer(-feature, names_to = "model", values_to = "rank") %>%
  dplyr::mutate(model   = factor(sub("_rank", "", model), levels = model_cols),
                feature = factor(feature, levels = rev(rank_tab$feature)))

ggplot(rank_long, aes(x = model, y = feature, fill = rank)) +
  geom_tile(colour = "white") +
  geom_text(aes(label = rank), size = 2.5) +
  scale_fill_gradient(low = "steelblue", high = "white") +
  labs(title = "Importance Rank by Model (ULB, Ordered by RSF)",
       x = "Model", y = "Predictor", fill = "Rank") +
  theme_bw()
```



4.3. Direction of Effects (LR and Cox)

```
## Step 1: Mean signed standardised coefficients with 95% split intervals.
## Positive = higher fraud odds (LR) or higher fraud hazard (Cox).
dir_tab <- explain_ulb$coef %>%
  dplyr::group_by(feature) %>%
  dplyr::summarise(
    LR_mean  = mean(LR),  LR_lo  = quantile(LR, 0.025),  LR_hi  = quantile(LR, 0.975),
    Cox_mean = mean(Cox), Cox_lo = quantile(Cox, 0.025), Cox_hi = quantile(Cox, 0.975),
    .groups = "drop"
  ) %>%
  dplyr::mutate(Same_direction = sign(LR_mean) == sign(Cox_mean)) %>%
  dplyr::arrange(dplyr::desc(abs(LR_mean)))

pander(dir_tab, digits = 3, split.table = Inf,
       caption = "Standardised Coefficients: LR vs Cox (Mean and 95% Split Interval)")
```

Table 5: Standardised Coefficients: LR vs Cox (Mean and 95% Split Interval)

feature	LR_mean	LR_lo	LR_hi	Cox_mean	Cox_lo	Cox_hi	Same_direction
V17	-4.11	-33.5	0.0104	-0.0125	-0.0466	0.0256	TRUE
V14	-3.36	-19.8	-0.894	-0.3	-0.341	-0.251	TRUE
V12	-2.81	-18.2	-0.368	0.0305	-0.033	0.0738	FALSE
V10	-2.68	-15.8	-0.711	-0.0791	-0.111	-0.0466	TRUE

feature	LR_mean	LR_lo	LR_hi	Cox_mean	Cox_lo	Cox_hi	Same_direction
V4	2	1.15	7.15	0.314	0.228	0.368	TRUE
V7	-1.79	-14.2	0.307	0.0741	0.0104	0.124	FALSE
V16	-1.73	-12.4	0.187	0.00965	-0.0176	0.0439	FALSE
V3	-1.38	-11	0.401	0.0308	-0.0211	0.0583	FALSE
V11	1.05	-0.0748	7.93	0.219	0.182	0.261	TRUE
V18	-0.897	-7.77	0.156	0.0461	0.0162	0.0792	FALSE
V2	0.742	-0.112	5.6	0.109	0.0685	0.146	TRUE
V9	-0.74	-5.06	0.203	-0.0335	-0.0633	0.00562	TRUE
Amount	0.592	-0.0292	2.57	0.0789	0.0357	0.125	TRUE
V6	-0.559	-2.5	-0.177	-0.0522	-0.101	-0.0216	TRUE
V22	0.43	0.184	0.961	-0.00505	-0.0493	0.0316	FALSE
V21	0.418	0.0209	1.93	0.0137	-0.00615	0.0323	TRUE
Time	-0.384	-0.743	-0.0845	-1.1	-1.18	-0.521	TRUE
V13	-0.381	-0.647	-0.196	-0.0726	-0.125	-0.017	TRUE
V19	0.306	-0.133	2.49	-0.0371	-0.0697	-0.00377	FALSE
V8	-0.277	-0.9	1.79	-0.0335	-0.0538	-0.00941	TRUE
V15	-0.23	-0.575	-0.0906	0.0608	0.016	0.104	FALSE
V20	-0.208	-0.513	0.569	0.0184	-0.0304	0.0623	FALSE
V26	-0.143	-0.277	0.0523	-0.131	-0.173	-0.0713	TRUE
V5	-0.126	-5.15	1.39	0.00954	-0.0365	0.047	FALSE
V27	-0.0819	-0.313	0.683	0.0814	0.0441	0.116	FALSE
V24	0.0718	-0.141	0.247	-0.0266	-0.0732	0.0149	FALSE
V25	0.0447	-0.147	0.285	0.103	0.0546	0.148	TRUE
V28	0.0253	-0.19	0.709	0.0429	0.00926	0.102	TRUE
V23	0.0108	-0.119	0.268	-0.056	-0.0935	-0.0183	FALSE
V1	3e-04	-3.93	1.5	-0.194	-0.239	-0.132	FALSE

```r
## Step 2: Share of predictors where LR and Cox agree on direction
pct_same_direction <- mean(dir_tab$Same_direction) * 100
pct_same_direction
```

```
## [1] 53.33333
```

Observations: `V14` is the one predictor all four models rank highly (first or second for LR, RF and RSF, and third for Cox), and `V4` ranks in the top six for all four. Beyond these, the models divide by the outcome they model rather than by model family: LR and RF agree closely (Spearman 0.873, with four shared top-five predictors), RSF agrees moderately with both (0.604 and 0.552), and Cox shows almost no agreement with any other model (-0.002 to 0.187). Raw `Time` illustrates the divide. It ranks first for Cox and third for RSF, but 20th for LR and 23rd for RF, so it matters only to the two models whose survival response it duplicates, consistent with the Time Sensitivity Analysis.

LR and Cox agree on the direction of only 53.3% of the coefficients, close to chance, although most disagreements involve small coefficients whose intervals span zero, and the strongest shared predictors (`V14`, `V10`, `V4`, `V11`) agree. The LR coefficient magnitudes should also be read with caution: `glm` reported fitted probabilities of 0 or 1 in 28 of the 30 splits, a sign of quasi-separation, and the very wide LR intervals (for example `V17`, -33.5 to 0.01) show that a few splits inflate the mean magnitudes. Cox's ridge coefficients are shrunk and dominated by `Time` (-1.10), which accounts for much of its disagreement with the other models.

5. Synthesis

The four models agree on the strongest ULB fraud signal, `V14` and to a lesser extent `V4`, but beyond that their explanations depend on the outcome being modelled. The two classification models agree closely, while the two survival models diverge from them, largely because they weight raw `Time` heavily as a duplicate of their

own response. Explanations of RSF on ULB should therefore be read as explanations of a survival model that partly encodes the proxy time ordering, rather than as a model-independent account of what drives fraud.